

1. STP (PVST) Priority

The default STP priority number is **32768**. It can be configured in increments of 4096 within a range of 0 to 61440, with lower values giving a switch a higher chance of becoming the **root bridge**. The priority combined with the system ID extension (VLAN ID) makes up the bridge priority.

- **Default Priority:** 32768
- **Range:** 0–61440 (multiples of 4096)
- **Lower = Better:** A lower value ensures the switch is preferred as the root bridge.
- **Root Primary:** 24576 or lower
- **Root Secondary:** 28672

Commonly Used Priority Values:

- **0:** Often used for the primary root bridge.
- **4096:** Often used for the secondary root bridge.
- **32768:** Default for all switches.

If multiple switches have the same priority, the one with the lowest MAC address becomes the root bridge.

2. HSRP Priority

- **Range:** 1 to 255.
- **Default:** 100.
- **Highest Wins:** A higher number is preferred; set to 255 for the primary router.
- **Preemption:** Use the standby preempt command to allow a router with a higher priority to take over, even if it comes online after a lower-priority router.
- **Configuration:** `standby [group-number] priority [priority-value]`